Editors' Suggestion

Optical absorption of subwavelength silicon nanoparticles at telecom wavelengths

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We investigate the heating of silicon nanoparticles due to absorption of light at 1550 nm wavelength. We conclude that the strong temperature dependence of the absorption processes in silicon leads to a thermal runaway effect, destroying the particle once it reaches a critical temperature under intense irradiation. Our results bear relevance for recent proposals to use silicon particles for levitodynamics experiments.

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I. INTRODUCTION

Levitodynamics harnesses levitated nanoparticles to address fundamental questions in physics and to develop new sensing paradigms [1,2]. A central goal of the field is to test quantum physics at increasingly large scales [3,4].

The workhorse of levitodynamics are optically levitated nanoparticles made of silica, the only material that withstands the heating due to intense laser irradiation in high vacuum. Those particles have allowed ground-state cooling of their center-of-mass motion [5–10], a prerequisite for progressing deeper into the quantum regime. However, a silica nanoparticle levitated in a typical optical trap in high vacuum is heated to around 1000 K by optical absorption [11,12]. This elevated temperature gives rise to blackbody radiation, which represents a source of decoherence for the mechanical motion of the particle [3,13,14].

Therefore, the levitodynamics community is searching for nanoparticles made of materials that outperform silica in terms of optical absorption. Reference [15] provides a table with investigated materials. Experimental evidence suggests that, under the intense optical irradiation present in an optical trap, no investigated material besides silica can tolerate pressures low enough to enter the regime of strongly underdamped motion required for sensing and quantum experiments.

As long as the surrounding gas pressure is high enough, cooling by the gas keeps the particle's internal temperature sufficiently low to allow for stable trapping [11,12]. However, below a critical pressure, the required balance between cooling (by gas and blackbody radiation) and heating (by optical absorption) leads to an equilibrium temperature high enough to destroy the particle.

In the search for suitable materials for future levitodynamics experiments, silicon nanoparticles [16] have emerged as promising candidates for trapping at telecom wavelengths around $\lambda = 1550$ nm [17,18]. Tabulated bulk properties raise hopes for lower absorption and higher refractive index in the near-infrared (NIR) wavelength range for silicon relative to silica [19]. Pioneering efforts have investigated rotational dynamics of optically trapped silicon nanoparticles [20]. Pressures in those experiments were limited to moderate vacuum around 1 mbar. While cavity cooling of silicon nanorods flying through an optical cavity in ultrahigh vacuum has been observed [21], their stable optical trapping at low pressures has not been reported. Regarding the absorption properties of silicon nanoparticles, a detailed study has been carried out for visible wavelengths [22]. An important next step is an experimental study of their optical absorption properties in the NIR.

In this work, we investigate the heating of silicon nanoparticles by optical absorption. We use Raman thermometry to measure the internal temperature of silicon nanoparticles under intense laser irradiation at 1550 nm. We find that the absorption of 1550 nm radiation by silicon nanoparticles increases strongly with temperature. This behavior is explained by a model based on previously reported optical properties of silicon. The temperature dependent absorption gives rise to a thermal runaway effect that leads to the destruction of silicon nanoparticles irradiated by telecom light once their internal temperature exceeds a threshold. Our study can explain why stable optical trapping of silicon nanoparticles in high vacuum has not been reported.

II. EXPERIMENTAL METHODS

The subject of our study are near-spherical silicon nanoparticles prepared according to Ref. [23]; see scanning electron microscope (SEM) image in inset of Fig. 1(a). Transmission electron microscopy shows high crystallinity of the silicon lattice (see Supplemental Material [24]). We suspend the silicon nanoparticles (diameters in the range $d_{\mathrm{NP}} \approx 220$–$400$ nm) in high purity ethanol and drop cast this solution onto a glass microscope coverslip. Our experimental setup is shown in Fig. 1(a). At its heart is an inverted optical microscope equipped with a reflective objective (Beck Optronic 5006, gold coated, numerical aperture NA = 0.65). Three lasers can be directed onto the sample. The first laser emits at the

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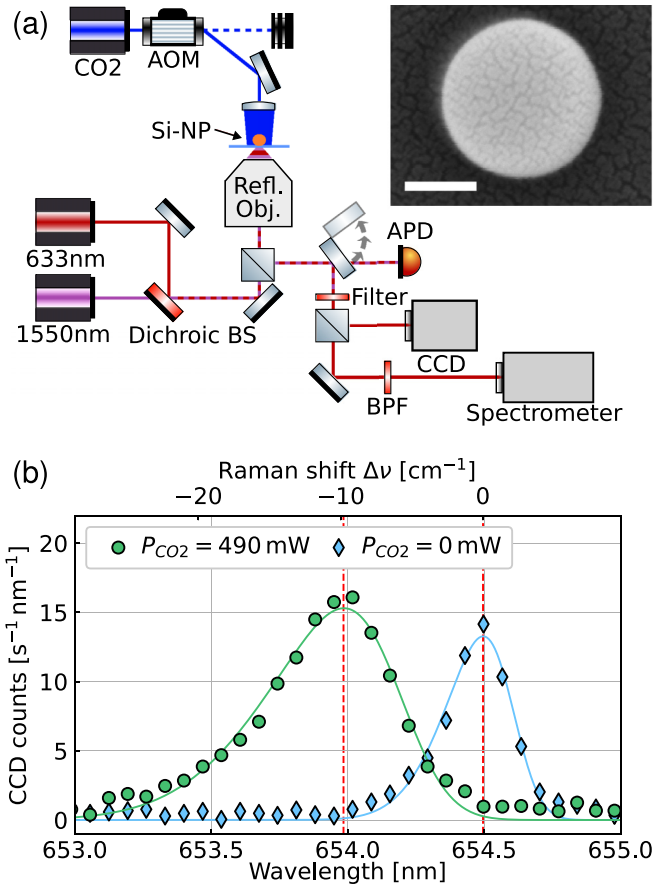


FIG. 1. (a) Experimental setup. We use an inverted microscope with a reflective objective to probe single silicon nanoparticles dispersed on a glass coverslip. The nanoparticle can be illuminated with a 633 nm Raman laser and a 1550 nm telecom laser. The Raman-scattered light can be analyzed with a spectrometer. A bandpass filter (BPF) suppresses all wavelengths except those close to the Raman band. An avalanche photodiode (APD) and a camera (CCD) serve for orientation on the substrate and alignment of the laser foci onto each other. The glass substrate can be heated with a CO_2 laser (wavelength 10.6 μm) from the top. Inset: scanning electron microscope (SEM) image of a typical silicon nanoparticle. For SEM imaging, a few-nm layer of platinum was sputtered onto the sample. The scale-bar indicates 100 nm. (b) Two exemplary Raman spectra collected on a representative silicon nanoparticle and their fits (see Supplemental Material [24] for the fit function). The blue diamonds represent the spectrum obtained with only the Raman laser guided at the sample. The green circles show the spectrum of the same silicon nanoparticle while the substrate is heated by the CO_2 laser. From the shift of the Raman spectrum by $\Delta\nu = -9.72$ cm^{-1}, we can extract the heating of the nanoparticle by $\Delta T = 414$ K above room temperature [25].

telecom wavelength 1550 nm. This "telecom laser" is focused onto the sample using the reflective objective. The telecom laser plays the role of the optical tweezer in a levitation experiment and the goal of this work is to investigate the heating of a silicon nanoparticle due to this laser. The second laser, in the following termed "Raman laser," emits at 633 nm and is also directed onto the sample through the objective. To this end, it is combined with the telecom laser using a dichroic beam splitter before entering the microscope. The third light source, in the following termed "heating laser," is a CO_2 laser emitting at 10.6 μm. It is focused from the top to a spot size on the sample of about 1 mm with a ZnSe lens (focal length 150 mm). The heating laser is strongly absorbed by the glass substrate which can thereby be heated up to ≈800 K. This heating can be controlled by tuning the heating laser power with an acousto-optic modulator (AOM).

We measure the internal temperature of the silicon nanoparticles by Raman thermometry [22,26]. To acquire a Raman spectrum, we collect the light scattered by a single silicon nanoparticle in the laser focus with the microscope objective and guide the scattered light to a spectrometer. To illustrate the concept of Raman thermometry, the Raman spectrum of a silicon nanoparticle (in the absence of the heating and the telecom laser, i.e., at room temperature) is shown as blue diamonds in Fig. 1(b). The power of the Raman laser is 10 μW and does not significantly heat the particle. The inelastically scattered photons have a wavelength centered around 654.5 nm, shifted from the laser, which emits at 633 nm. The shift corresponds to 519 cm^{-1}—the Raman shift of bulk silicon at room temperature [27]. The green circles in Fig. 1(b) show the Raman spectrum in the presence of the heating laser (power 490 mW). We observe that, in the presence of the heating laser, the Raman shift is reduced compared to the unheated case by $\Delta\nu = -9.72$ cm^{-1}. In silicon, the Raman shift $\Delta\nu$ with temperature T is described by Balkanski's formula, a monotonic, invertible function, making it possible to infer the sample temperature from a given shift of the Raman line [25]. According to this method, the shifted spectrum in the presence of the heating laser indicates a nanoparticle temperature of 414 K above room temperature (714 K absolute temperature; see Supplemental Material [24] for details on the analysis).

III. RESULTS

We start by analyzing the heating of a silicon nanoparticle by the telecom laser. To this end, we monitor the nanoparticle temperature change ΔT as a function of telecom laser power P_{1550}. Here, ΔT is the temperature change relative to the nanoparticle temperature in the absence of the telecom laser, which is room temperature (300 K) in the first measurement series. The power of the Raman laser is kept fixed at $P_{633} = 10$ μW. In Fig. 2, we show the temperature change ΔT as a function of P_{1550} as green circles for a nanoparticle labeled NP1. This nanoparticle has a diameter of 230 nm (see SEM image in Supplemental Material [24]). We observe that the temperature changes by no more than 10 K due to irradiation with the telecom laser. We thus conclude that, for a silicon nanoparticle on a substrate, no significant heating is induced by irradiation with a telecom laser with more than 1 W of power focused to a diffraction limited spot. The full width at half maximum for the 1550 nm laser focus is measured to be 1100 nm; see Supplemental Material [24]. A power of $P_{1550} = 100$ mW thus corresponds to an intensity of $I = 73$ mW/μm^2. We show a similar behavior for two further nanoparticles (NP2 and NP3, data shown as blue and orange circles in Fig. 2, respectively; SEM images in Supplemental Material [24]).

Next, we change the temperature of the substrate, and thereby of the silicon nanoparticle under investigation, to

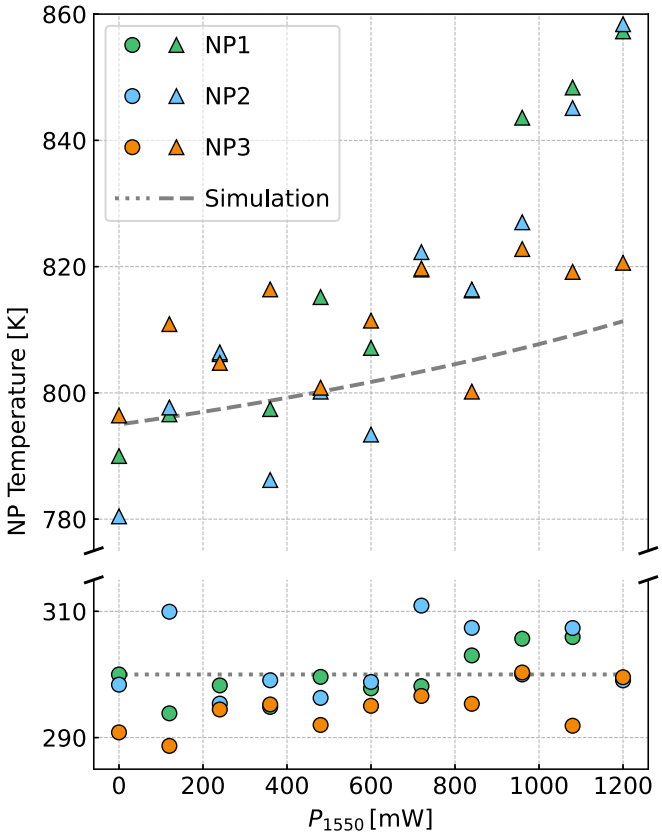


FIG. 2. Internal temperature for three different silicon nanoparticles (labeled NP1, NP2, and NP3), as a function of telecom laser power P_{1550}. The measurements are carried out at two different substrate temperatures: $T_{\mathrm{env}} = 300$ K ($P_{\mathrm{CO2}} = 0$ mW, circles) and $T_{\mathrm{env}} = 790$ K ($P_{\mathrm{CO2}} = 490$ mW, triangles). At elevated substrate temperatures, we observe significant heating by the telecom laser. In contrast, this optical heating is absent for the unheated substrate. The dashed (dotted) line is a calculation based on the theory outlined in the Supplemental Material [24] assuming 795 K (300 K) environmental temperature and a particle diameter of 260 nm. The figure is reproduced including error bars in the Supplemental Material [24].

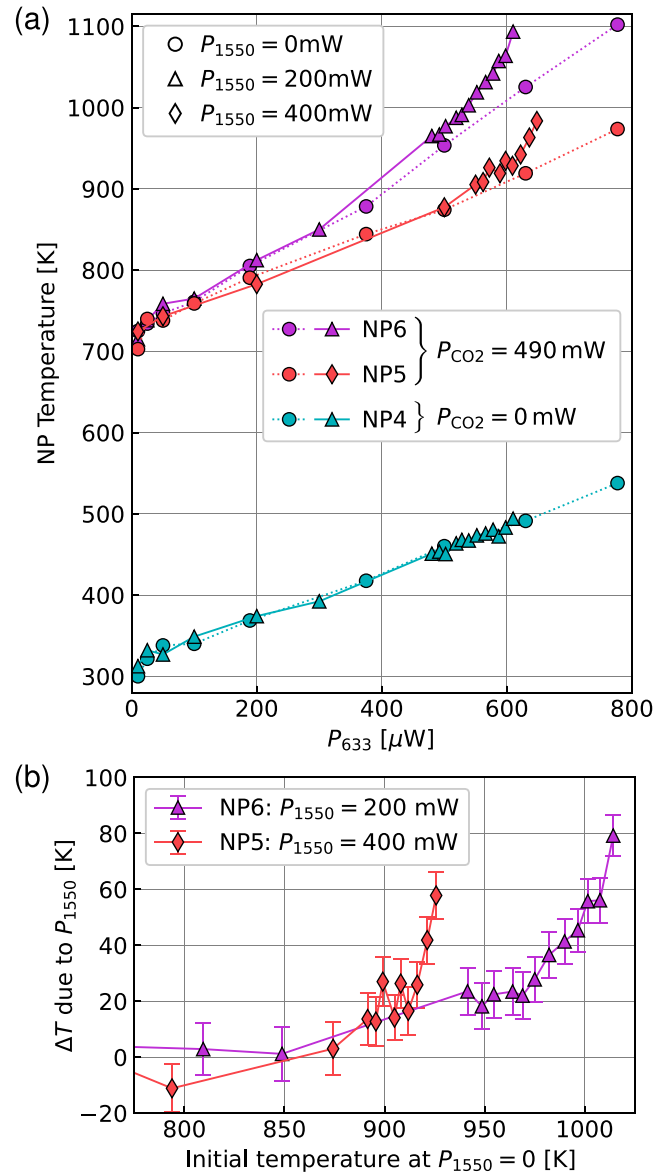


FIG. 3. (a) Internal temperature for three different silicon nanoparticles (in turquoise, red, and purple) as a function of the Raman laser power P_{633}. Triangles and diamonds (circles) are in the presence (absence) of telecom laser irradiation. The turquoise data set (NP4) is measured in the absence of the heating laser; the red and purple data sets are measured under heating with power $P_{\mathrm{CO2}} = 490$ mW. (b) Temperature change ΔT due to the telecom laser as a function of temperature for NP5 and NP6, extracted from data shown in (a). The temperatures at $P_{1550} = 0$ are extracted by linear interpolation between the measured data points (circles). Error bars denote one standard deviation.

$T_{\mathrm{env}} = 790$ K by irradiating it with the heating laser using a power $P_{\mathrm{CO2}} = 490$ mW. We again measure the additional heating generated by the telecom laser, shown as the green triangles for NP1 in Fig. 2. We see an increase in the nanoparticle's internal temperature by $\Delta T = 68$ K for the maximally available telecom-laser power $P_{1550} = 1.2$ W.

The observations in Fig. 2 indicate that heating of a silicon nanoparticle due to absorption at telecom wavelengths depends on the temperature of the nanoparticle. More specifically, absorption and associated heating become more severe at elevated temperatures. This behavior is observed consistently also for two additional particles, labeled NP2 (blue data points) and NP3 (orange data points) in Fig. 2.

Investigating the temperature dependent absorption of our silicon particles further requires pushing their internal temperature higher. For practical reasons, this can neither be achieved with larger telecom-laser powers (due to the limited output power of our source) nor via stronger heating by the CO_2 laser (due to deformation of the coverslip beyond 800 K). Therefore, we resort to using the (above band gap) absorption of the Raman laser as a means to tune the nanoparticle temperature.

Figure 3(a) shows as turquoise circles the measured internal temperatures of another silicon nanoparticle (labeled NP4) as a function of Raman-laser power P_{633} in the absence of both heating and telecom laser light. We observe a linear dependence of nanoparticle temperature as a function of

P_{633}. When repeating the measurement, still in the absence of a heating laser, but in the presence of $P_{1550} = 200$ mW of telecom light (turquoise triangles), no appreciable difference of the temperature as a function of P_{633} is observed. We conclude that, at temperatures up to ≈500 K, the heating by 200 mW of telecom light is irrelevant when compared to the heating caused by $P_{633} = 800$ μW of visible Raman laser light, which amounts to 200 K. This observation makes sense when taking into account that the photon energy of the Raman laser exceeds the band gap of silicon (such that it is strongly absorbed), while the telecom laser's photon energy is below band gap.

The situation is strikingly different for a nanoparticle heated by the heating laser ($P_{CO2} = 200$ mW) to 730 K [NP5, red circles and diamonds in Fig. 3(a)]. When probing the absorption of the visible Raman laser in the absence of the telecom laser (red circles), we still find a linear rise in nanoparticle temperature with P_{633}. However, in the presence of telecom light (power $P_{1550} = 200$ mW), the nanoparticle temperature grows nonlinearly with P_{633} (red diamonds). In other words, the temperature increase due to a fixed amount of telecom light is larger at higher temperatures as compared to lower temperatures. Thus the absorption coefficient at 1550 nm increases with temperature. We stress that this nonlinear growth *cannot* be due to a nonlinear or temperature dependent absorption of the visible Raman laser, since this is ruled out by the linear dependence of T on P_{633} in the absence of the telecom laser [red circles in Fig. 3(a)]. Our observations in Fig. 3(a) are therefore in line with those in Fig. 2. Both suggest that absorption of telecom light in silicon nanoparticles becomes stronger as their temperature grows. To confirm our observation, Fig. 3(a) contains data collected on another nanoparticle (NP6, purple data points), showing a qualitatively similar behavior and confirming our conclusions. For this data set, the telecom power used is $P_{1550} = 200$ mW.

We stress that the measurements in Fig. 3(a) destroyed nanoparticles NP5 and NP6. To be concrete, for each nanoparticle, we first perform the sweep of P_{633} in the absence of the telecom laser (circles). We then return to low P_{633} and increase it in the presence of the telecom laser (triangles and diamonds). The rightmost diamond (for NP5) and the rightmost triangle (for NP6) are the last data points collected before the respective particle disintegrates upon increasing P_{633} one more step. We confirm particle disintegration using SEM images (see Supplemental Material [24]).

The nonlinear trend of the IR absorption coefficient observed in Fig. 3(a) can be more easily visualized in terms of the temperature differentials induced by the 1550 nm laser. In Fig. 3(b), we replot the data in Fig. 3(a), showing the temperature increase ΔT due to the telecom laser as a function of nanoparticle temperature in the absence of the telecom laser (set by the heating and the Raman laser powers). As an example, when NP6 is heated to 1000 K by 590 μW of Raman laser and $P_{CO2} = 490$ mW of heating laser, adding $P_{1550} = 200$ mW of telecom laser increases the temperature by another $\Delta T = 55$ K to 1055 K. We observe that a fixed telecom laser power (200 mW for NP6 shown in purple, 400 mW for NP5 shown in red) gives rise to more heating the hotter the particle already is. We therefore conjecture that our particles are destroyed under too intense optical radiation by an effect termed "thermal runaway," which has been described in silicon photonic structures [28]. Thermal runaway is a process where optical absorption leads to heating, which increases absorption, leading to positive feedback that spirals sample temperature out of control. In the following, we present a theoretical analysis corroborating our conjecture.

IV. MODEL

To explain our observations, we turn our attention to the absorption of silicon at 1550 nm. Following the model developed in Ref. [28], we express the total absorption coefficient of silicon α_{tot} as the sum of three contributions:

$$\alpha_{tot}(T, I) \approx \alpha_0 + \alpha_{TPA}(T, I) + \alpha_{FC}(T) + \alpha_{BG}(T). \quad (1)$$

The first contribution α_0 is the measured absorption of silicon at room temperature for vanishingly low intensity. We use the value $\alpha_0 = 4.3 \times 10^{-6}$ cm^{-1} [18]. The second term accounts for two-photon absorption (TPA) via the TPA coefficient α_{TPA}, which depends on both temperature T and light intensity I. Specifically, we use $\alpha_{TPA} = \beta_{TPA}(T)I$ following Ref. [29], from which we extract β_{TPA}. We note that, as will become clear, throughout this work α_{TPA} largely exceeds α_0 at telecom wavelengths. The third contribution to α_{tot} is due to intraband free-carrier absorption, described by the temperature dependent coefficient α_{FC} found in Ref. [30]. The final contribution due to interband (band gap) absorption is described by the coefficient α_{BG}, whose temperature dependence we take from Ref. [30]. The temperature dependence of band gap absorption can be understood as the thermoelastic effect expanding the lattice structure as a result of heating, which in turn reduces the band gap, leading to increased absorption. For details on the absorption processes in silicon and an overview on the available data, see Ref. [28].

The resulting total absorption coefficient $\alpha_{tot}(T, I)$ for silicon at 1550 nm is shown as a function of temperature in Fig. 4 as solid lines for two different light intensities typically used in optical trapping experiments ($I = 0.1$ W/μm^2 in orange and $I = 1$ W/μm^2 in purple). We also plot the contributions to $\alpha_{tot}(T, I)$ from free-carrier absorption α_{FC} and from interband (band gap) absorption α_{BG} as dashed lines; see legend. In addition, as the red solid line, we plot the linear absorption coefficient α_0. We note that, for typical trapping intensities and above room temperature, linear absorption of silicon provides a negligible contribution to α_{tot}. At temperatures below 500 K, absorption is dominated by TPA. TPA has a relatively weak temperature dependence, increasing by less than an order of magnitude between 400 and 1600 K. For $I = 0.1$ W/μm^2, free carrier absorption α_{FC}, which strongly grows with increasing temperature, overtakes TPA above temperatures around 600 K. For $I = 1$ W/μm^2, free-carrier absorption dominates over TPA above 700 K. Our model also shows that band gap absorption α_{BG} sharply rises above 1050 K and becomes comparable to the otherwise dominant free-carrier absorption at temperatures above 1500 K, close to the melting point of silicon at 1680 K.

Equipped with the total absorption coefficient of silicon, we turn our attention to modeling our experiment. We

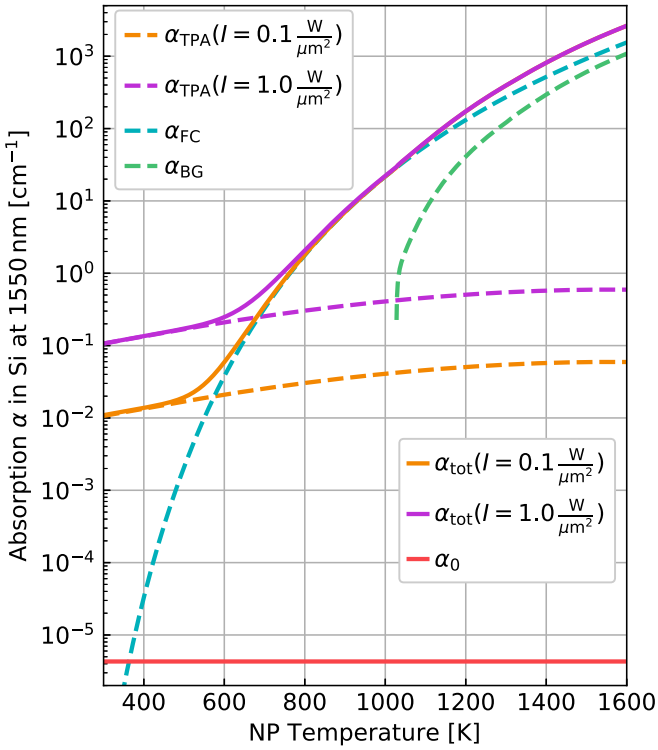


FIG. 4. Modeled absorption coefficient for silicon $\alpha_{tot}(T, I)$ at 1550 nm as a function of temperature T and its contributions according to Eq. (1) for two different intensities.

aim to find the internal temperature of the silicon nanoparticles given the laser intensity they are exposed to. To this end, we seek the internal temperature T for which the balance is struck between absorbed and emitted power according to

$$P_{abs}(T, I) = P_{cond}(T, T_{env}) + P_{BB}(T, T_{env}). \quad (2)$$

Here, P_{abs} is the power absorbed in the nanoparticle, P_{cond} is the power emitted by the nanoparticle through conductive heat transfer (through the air or the glass coverslip), and P_{BB} is the power emitted through blackbody radiation.

For silicon nanoparticles deposited on a glass coverslip, we approximate the conductive heat transfer via a simple model employing a heat transfer coefficient h according to [26]

$$P_{cond}(T) = hA_{NP}(T - T_{env}), \quad (3)$$

where A_{NP} is the surface area of the silicon nanoparticle. Furthermore, we make the assumption $P_{cond} \gg P_{BB}$, which is well justified for our experimental parameters. We extract the proportionality constant h from Fig. 6 in [26]. We solve Eq. (2) numerically for T, while calculating the absorbed power using our model for $\alpha_{tot}(T, I)$ in Eq. (1) and Mie-Theory for a free nanoparticle provided in Ref. [31]. The solutions of this simple model are shown in Fig. 2 as the dashed and dotted lines for the corresponding experimental parameters. We note that the experimentally observed heating for NP1 and NP2 in the presence of the heating laser is about twice larger than the calculated result. To put this into perspective, we stress that the theory curves in Fig. 2 are not fits but calculations using literature values together with measured experimental parameters. The results of these calculations depend strongly on the absorbed power P_{abs} (which in turn is strongly dependent on particle size and the local intensity) as well as the cooling coefficient h. For example, increasing P_{abs} by a factor of 2 would already yield a temperature increase of $\Delta T \approx 52$ K at $P_{1550} = 1200$ mW, matching the experimental observation. Therefore, while our calculation does not provide a perfect match with the experiment, it captures the salient feature of our observation, which is the increasing absorption of silicon nanoparticles with increasing temperature.

V. IMPLICATIONS FOR TRAPPING OF SILICON NANOPARTICLES IN VACUUM

In this section, we discuss the consequences of optical absorption on the prospects of using silicon nanoparticles in levitodynamics experiments. Such experiments in the quantum regime require ultrahigh vacuum, where the damping of the mechanical motion of the particle in the optical trap is essentially absent [1].

We start this section by presenting experimental data on optical trapping of silicon nanoparticles in vacuum. The particles are the same batch as those used throughout the rest of this manuscript. We dissolve the particles in ethanol and load them with a nebulizer into an optical tweezer trap, sketched in Fig. 5(a), formed by a lens with NA = 0.7, focusing a laser beam (wavelength 1550 nm) to a diffraction limited spot. We confirm the presence of the particle in the trap by detecting the scattered field on a photodetector. We load the trap at atmospheric pressure and at a given laser power and then slowly evacuate the vacuum chamber. We consistently observe that the particle disappears from the trap during pump down and we record the critical pressure at which particle loss occurs. We repeat this experiment at different optical powers in the laser beam forming the tweezer P_{tw}. The observed critical pressures are plotted as a function of P_{tw} in Fig. 5. We observe that the critical pressure where particle loss occurs systematically shifts towards larger values with increasing tweezer power.

To explain the observed behavior, we return to the power balance in Eq. (2). In high vacuum, cooling by blackbody radiation dominates over conductive cooling, such that $P_{BB} \gg P_{cond}$. Accordingly, in high vacuum, blackbody emission must compensate absorptive heating to guarantee a finite, stable internal temperature of the nanoparticle. To check whether such an equilibrium temperature exists in the blackbody dominated regime, we calculate the emitted blackbody power P_{BB} as a function of particle temperature and plot it in Fig. 6 as the blue line, assuming an environment temperature of 300 K (calculation detailed in Supplemental Material [24]). In addition, we plot the absorbed power P_{abs} for an illumination intensity due to the telecom laser of $I = 0.1$ W/μm^2 in orange. We note that, for the entire temperature range from room temperature up to 1600 K, the absorbed optical power is always at least five orders of magnitude larger than the power emitted via blackbody radiation. In other words, for typical optical intensities used for trapping in focused fields, our model predicts that no stable temperature exists for silicon nanoparticles in high vacuum.

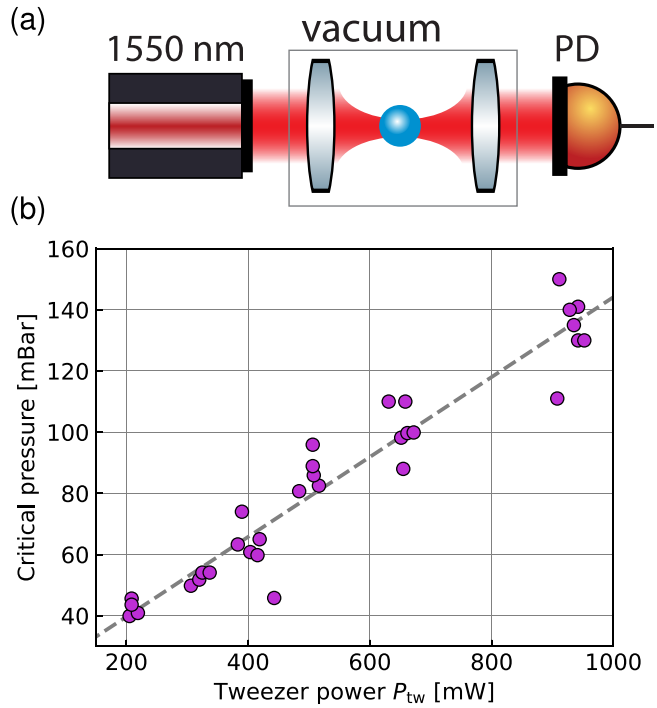


FIG. 5. (a) Optical trapping setup. A laser (1550 nm) is focused with an aspheric lens inside a vacuum chamber to form an optical trap. The light scattered by the trapped particle is directed to a photodetector (PD). (b) Critical pressure where particle loss occurs during pump down as a function of tweezer power P_{tw}. Each data point represents one silicon nanoparticle lost from the optical trap.

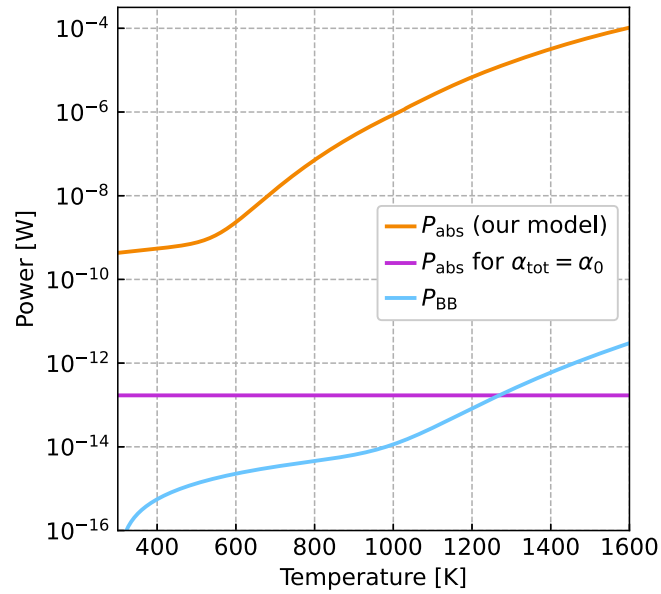


FIG. 6. Calculated absorbed power P_{abs} (orange line) as a function of temperature using our model for α_{tot} as shown in Fig. 4 for a nanoparticle with diameter $d_{NP} = 260$ nm irradiated with an intensity of $I = 0.1$ W μm^{-2}. The radiated blackbody power P_{BB} for a particle of the same size is shown in blue as a function of temperature. We find that P_{abs} and P_{BB} are off by five to six orders of magnitude for any temperature T such that no equilibrium temperature exists. The purple line shows the absorbed power only due to $\alpha_0 = 4.3 \times 10^{-6}$ cm^{-1} (neglecting TPA, band gap absorption, and free-carrier absorption).

The findings in Fig. 6 can explain the particle loss we observe experimentally in Fig. 5 when decreasing the pressure for the trapped silicon nanoparticles. As the pressure decreases, P_{cond} decreases, P_{BB} is insufficient to compensate P_{abs} in Eq. (2), and the nanoparticle will heat up. The temperature dependence of $\alpha_{tot}(T)$ leads to even higher absorption, boosting the nanoparticle temperature even further. This self-enhancing *thermal runaway* eventually leads to the loss of the nanoparticle from the trap [28]. This thermal runaway can be very rapid due to the high free-carrier and band gap absorption at elevated temperatures (see Fig. 4). The thermal runaway can thus explain both the particle loss from the optical trap observed in Fig. 5 and destruction of the nanoparticles on substrate, as observed in Fig. 3. Improvements in material purity could allow for shifting the onset of the thermal runaway to larger optical intensities.

Note that when considering the absorbed power using intrinsic absorption at room temperature α_0 only (purple line in Fig. 6), one concludes that there exists an equilibrium temperature just below 1300 K at an irradiation intensity of $I = 0.1$ W μm^{-2} (crossing of blue and purple lines in Fig. 6) [18]. However, taking into account the contributions of TPA and band gap absorption, our model suggests that the intensity of the 1550 nm trapping beam would have to be as low as $I \approx 5$ μW μm^{-2} for successful high-vacuum trapping of a silicon nanoparticle with diameter $d_{NP} = 260$ nm in a laser focus. Thermal stability therefore requires dealing with an extremely shallow optical trap depth. While our experimental findings are well described by our model, the investigation of larger particles reaching well beyond the dipolar approximation remains a relevant endeavor. Indeed, a way to circumvent large field intensities could be to use Mie resonances of nanoparticles for trapping at intensity minima [18]. Notably, even for this low-field configuration, it has been calculated that for silicon the contribution to absorption by α_0 already leads to internal temperatures beyond 1000 K in high vacuum for typical laser intensities [18], where α_0 underestimates the absorption coefficient by more than five orders of magnitude (see Fig. 4). For purely optical trapping experiments with small particles, our results therefore suggest that silicon may not be a straightforward choice of material. Nevertheless, the prospect of harnessing Mie resonances of silicon nanoparticles remains enticing [18]. Nonoptical suspension mechanisms, such as radiofrequency traps [32,33], could allow for the use of sufficiently weak laser intensities to harness the rich optomechanical interaction offered by resonant silicon nanoparticles while circumventing the absorption linked to large optical intensities.

VI. CONCLUSION

In conclusion, we have experimentally investigated the absorption of light at 1550 nm wavelength by silicon nanoparticles. We have observed that the absorption of telecom light by these particles increases with increasing particle temperature. Furthermore, we have developed a model that describes

heating of dielectric particles by optical absorption (using Mie theory), radiative cooling by blackbody emission (using dipolar scattering theory), and conductive cooling. In addition, we have compiled literature values for the absorption properties of silicon. We have found that, for light intensities typically used for optical levitation experiments, and at temperatures between 300 K and 600 K, absorption is dominated by two-photon absorption. Above 600 K, free-carrier absorption becomes dominant and becomes stronger with increasing temperature. This temperature dependence can lead to a thermal runaway effect when cooling of the particle is insufficient to balance the optical heating. We have experimentally characterized the onset of this thermal runaway effect for particles prepared on a substrate. We have furthermore demonstrated the loss of silicon nanoparticles from an optical trap under vacuum. This loss can also be explained by the thermal runaway caused by the temperature dependence of the absorption. Our results suggest careful consideration of the optical properties of silicon in proposals that involve optical trapping of silicon nanoparticles in levitodynamics experiments in vacuum.

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DATA AVAILABILITY

The data that support the findings of this article are openly available [34].

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